Jitter Tolerance Theory and Characterization Procedure

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4. Jitter Tolerance

4.1. Conceptual Overview

Jitter Tolerance is a test to quantify the ability of a receiver to interpret and recover a jittered signal. This usually involves a clock-data recovery circuit (CDR) that can track jitter applied to a data stream (or clock).

CDR tracking is impacted by the bandwidth applied at the Error Detector receiver on the M8020A J-BERT High-Performance BERT. A reduction in CDR bandwidth will shorten and/or shift the tracking region of the Jitter Tolerance profile in the frequency domain.

Jitter Tolerance margin is also impacted by amplitude and frequency relax times and bit error rate. Amplitude relax time adjusts the settling time when the magnitude of the applied jitter is decreased or increased. Frequency relax time adjusts the settling time when the frequency of the applied jitter is decreased or increased.

Bit error rate is the ratio of the number of bit errors over the total number of bits transmitted. Jitter Tolerance standards traditionally dictate BER specifications of 1E-12, 1E-15, or 1E-17. As electrical testing transitions from NRZ to PAM-X these BERs cannot be achieved by the receiver without FEC (forward error correction) so the requirements are relaxed to a BER of approximately 1E-5.

4.2. Signal Paths and Test Points

Note: Although the scope of this paper is limited to specific test equipment, signal paths and test points are dictated by standards specifications in addition to instrument and DUT circuitry. Functional similarities across all DUTs and TE allow the following description to be abstracted to apply to a generalized Jitter Tolerance configuration, independent of DUT type and instrument models.

Use of the M8020A J-BERT High-Performance BERT Error Detector requires that the signal as recovered by the DUT receiver be re-transmitted via far-end loopback without attenuation or other distortion to a secondary receiver at the Error Detector. This presents challenges introduced by clock-data alignment, which is the synchronization of the clock and data signals.

It is also critical to adjust the M8020A J-BERT High-Performance BERT receiver and CDR settings so as not to filter out jitter or other attributes seen by the DUT. The Error Detector CDR bandwidth should exceed that of the DUT such that all transmitted jitter is observed by the Error Detector. If a particular standard or application specifies a receiver and CDR configuration this must be implemented at the DUT, not at the Error Detector. If the RX/CDR configuration is implemented at the DUT the DUT RX/CDR is characterized; if implemented at the Error Detector the characterization applies to the cascaded signal path of DUT RX/CDR and Error Detector RX/CDR.

4.3. Parameters and Concepts

The following subsections address key Jitter Tolerance concepts and implementation caveats that factor into the appropriateness of a test configuration for a given application. Plots have been generated to discuss how Jitter Tolerance results should be interpreted and how parameters and controls impact performance. Screen captures are also provided to guide the GUI configuration process as detailed in Section 2.

Jitter Tolerance is a test to characterize a receiver's performance and capabilities. Parameters associated with transmitter equalization and swing and receiver circuitry (CDR, DFE, and CTLE settings, etc.) all change the shape of the resultant signal and impact margin.

NOTE: The lab equipment available for generating the data presented in this paper did not include an M8061A Multiplexer 2:1 With De-Emphasis 32 Gb/s or an N4877A Clock Data Recovery & Demultiplexer. All subsequent data and screen captures were generated with the Jitter Tolerance configuration for data rates below the maximum allowable M8020A J-BERT High-Performance BERT bandwidth of 16Gb/s as detailed in Section 2.

For all test cases in this discussion the DUT is the JBERT RX. There is no device in the path between the JBERT Pattern Generator and Error Detector and the test is measuring JTOL of the JBERT transmitter at the JBERT receiver when subject to various test conditions. The implications of having a DUT present in the system path are explored in depth where appropriate.

4.3.1 Reference Clock

A standard Jitter Tolerance configuration utilizes a reference clock during data transmission. The reference clock supplied to the DUT is responsible for clocking the data into the receiver at the correct rate.

The reference clock PLL cleans up the clock signal that is transmitted to the reference clock input of the DUT. It is critical to verify that the input reference clock is clean and stable. To perform Jitter Tolerance testing at the M8020A J-BERT High-Performance BERT Error Detector the signal as interpreted at the DUT RX must be retransmitted to the Error Detector via loopback, and with as little distortion and attenuation as possible. This retransmitted data stream is recovered via the reference clock signal that is retimed by the DUT PLL at the reference clock input.

Jitter on the input reference clock can distort the clock signal such that it is not ideal. Other timing artifacts can also impair the clock signal, such as those which result in DCD.

Jitter on the reference clock uncorrelated to the jitter on the data stream will close the eye. This will result in errors when evaluating the transmitted bits at the M8020A J-BERT High-Performance BERT Error Detector. The pattern specified at the Error Detector will not match the bits of the incoming data stream due to a bit value of 0 being mistakenly interpreted as a 1, and vice versa.

Jitter on the reference clock that is synchronous in frequency and magnitude to the jitter on the data stream will artificially improve the measure of CDR tracking capability. The signal clocking the data stream will be aligned with the jittered data and conceal real timing impairments. In terms of implementation this translates to fewer errors when feeding the measured data into the Error Detector and matching the binary digital interpretation to a predetermined and expected pattern. If there is no jitter on the reference clock then a Jitter Tolerance test accurately characterizes receiver performance subject to particular impairments and TX, RX and CDR settings.

4.3.2 Jitter Tolerance Tracking Regions

Effectively, a Jitter Tolerance test will shift a data signal away from its ideal timing location by applying Sinusoidal Jitter of a particular frequency and magnitude. When the magnitude of the SJ at this frequency is large enough the receiver will begin to incur errors when interpreting signal location with respect to the predetermined reference threshold and digitizing the incoming data stream. In other words, the binary pattern of the received signal will not match the expected pattern in memory.

Different frequency regions within the Jitter Tolerance frequency sweep are subject to different tracking capabilities by the receiver under test. Here it is important to note that the specific frequencies denoting the start and end of each region depend on the incoming signal and receiver characteristics. Data rate, TX and RX settings and RX CDR parameters play a huge role in determining which frequencies can be completely tracked by the DUT, which can only be partially tracked, and which can never be tracked.

The following screen capture clearly shows the three different frequency sweep regions corresponding to the low, mid, and high frequency sweep ranges specific to this particular test. These regions will shift and expand or contract in accordance with test-specific parameters such as data rate and CDR selections. In the low frequency sweep region, the jitter on the incoming data stream is easily tracked, in the mid frequency sweep region the jitter is tracked with some difficulty, and in the high frequency sweep region the jitter is not tracked at all. The jitter profile of Figure 21 corresponds to a large eye and well-behaved signal subject to minimal peaking and jitter amplification artifacts.

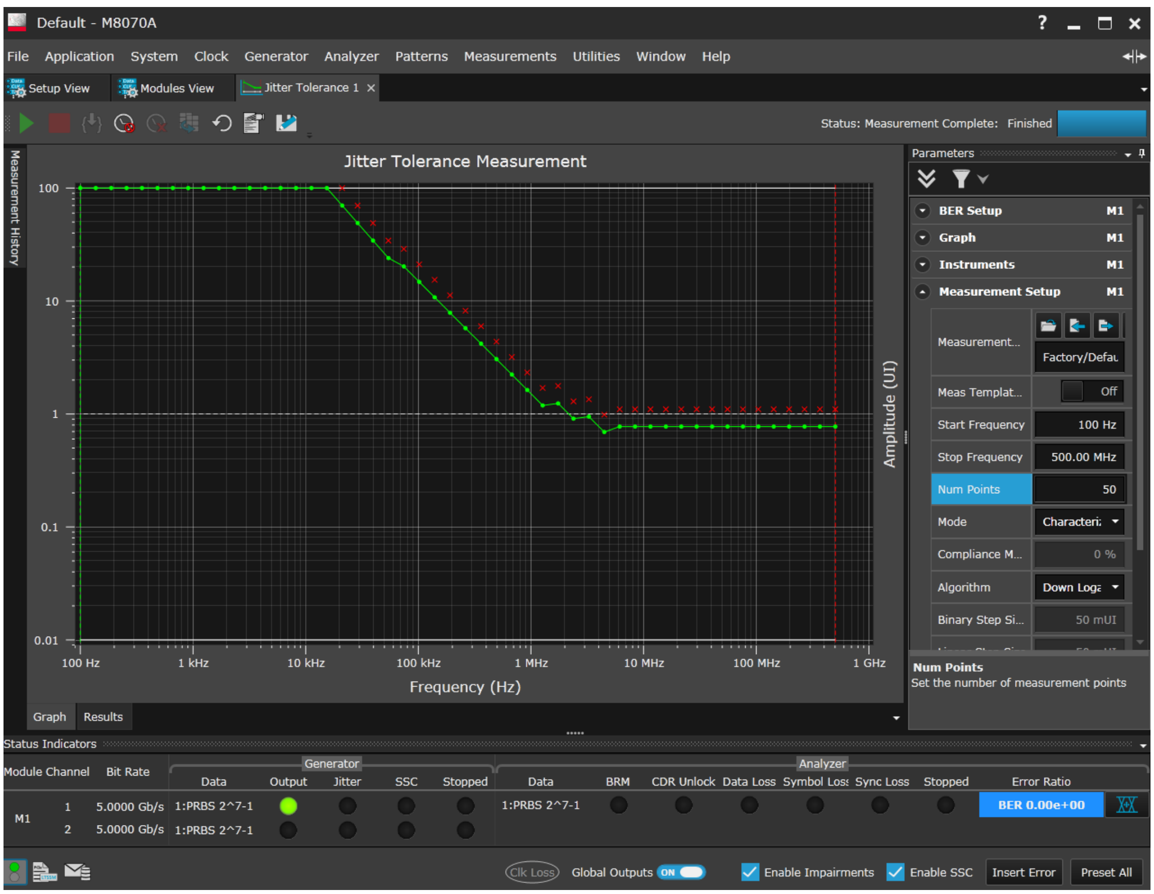

Figure 21 Jitter tolerance test results as collected and reported by the JBERT Error Detector. The test environment was minimally stressed and configured to generate a large, clean eye whose JTOL profile is familiar and mimics the ideal tracking behavior and SJ masks found in standards specifications and JTOL literature. The low, mid, and high frequency sweep regions are clearly visible and correspond to the three curves of different slope. Data Rate: 5Gb/s; BER: 1E-9; Search Algorithm: Down Logarithmic; Log Step Size: 30%; CDR Loop Order: 1st; CDR Loop Bandwidth: 2MHz; Jitter: none; TX Emphasis: none; Pattern: PRBS7; SJ Frequency Sweep Range: 100kHz–100MHz.